tests are carried out to determine the physical, chemical and bacteriological impurities present in water.

1. Physical tests
2. Chemical tests
3. Bacteriological tests

The list of tests and properties/ Impurities determined in the test are tabulated below:

1. Physical Tests

i. Turbidity

Turbidity is caused by the presence of finely divided, suspended and colloidal matters like clay, loam and sand or micro-organisms. It is more during floods. It is a measure of its resistance to the passage of light through it. It is expressed in parts per million(ppm) or milligram per litre(mg/l). The standard unit of turbidity is the turbidity produced by one part of finely divided silica (Fuller's earth) in a million parts of distilled water. For drinking water, the permissible turbidity is 5 to 10ppm. It is measured by Jackson Turbidimeter.

ii. Colour

Dissolved organic matter from decaying vegetation or some inorganic materials such as coloured soils, etc., may import colour to the water. The test for true colour should be taken up only removing all the suspended particles by centrifuging. The sample is then compared for colour with standard colour solutions or colour discs.

The unit of colour is the colour produced by one milligram of platinum cobalt in a litre of distilled water. The colour of public water supply should not exceed 20 mg/s. The colour is measured by Tintometer.

iii. Taste and Odour

The dissolved organic materials or the inorganic salts or the dissolved gases may impart tastes and odours to the water. Taste and odour are generally combined together. Tastes may be sweet, bitter, salty brackish and irritating. Odours may be fishy, earthy, grassy, mouldy, vegetable, etc. The odour is identified by using osmoscope.

2. Chemical Tests

i. Total solids

The total solids consist of dissolved and suspended solids. The total solids present in water can be determined by evaporating a sample of water and weighing the dry residue left. The suspended solids can be found by filtering water sample and weighing the residue left on the filter paper. The difference between the total solids and suspended solids will give the dissolved solids. The amount of total solids should preferably be less than 500 ppm and should never exceed 1000 ppm in any case.

ii. Hardness

It is the property of water which prevents the lathering of soap. It is due to the presence of bi carbonates of calcium and magnesium and is termed as temporary hardness or carbonate hardness. It can be removed either by boiling or by adding lime to water. The permanent hardness or non carbonate hardness is due to sulphates and chlorides of calcium and magnesium. It cannot be removed by boiling. It requires water softening.

iii. pH value

The pH value of water indicates the logarithm of reciprocal of hydrogen ion concentration present in water. It is thus an indicator of the acidity or alkalinity of water. The pure water consists of positively charged(H+)Hydrogen ions combined equally with negatively charged Hydroxyl (OH−) ions. The water is said to be acidic when H+ ions are excess than OH− ions and the pH value ranges from 0 to 7. The

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